Sri Lanka Institute of Information Technology

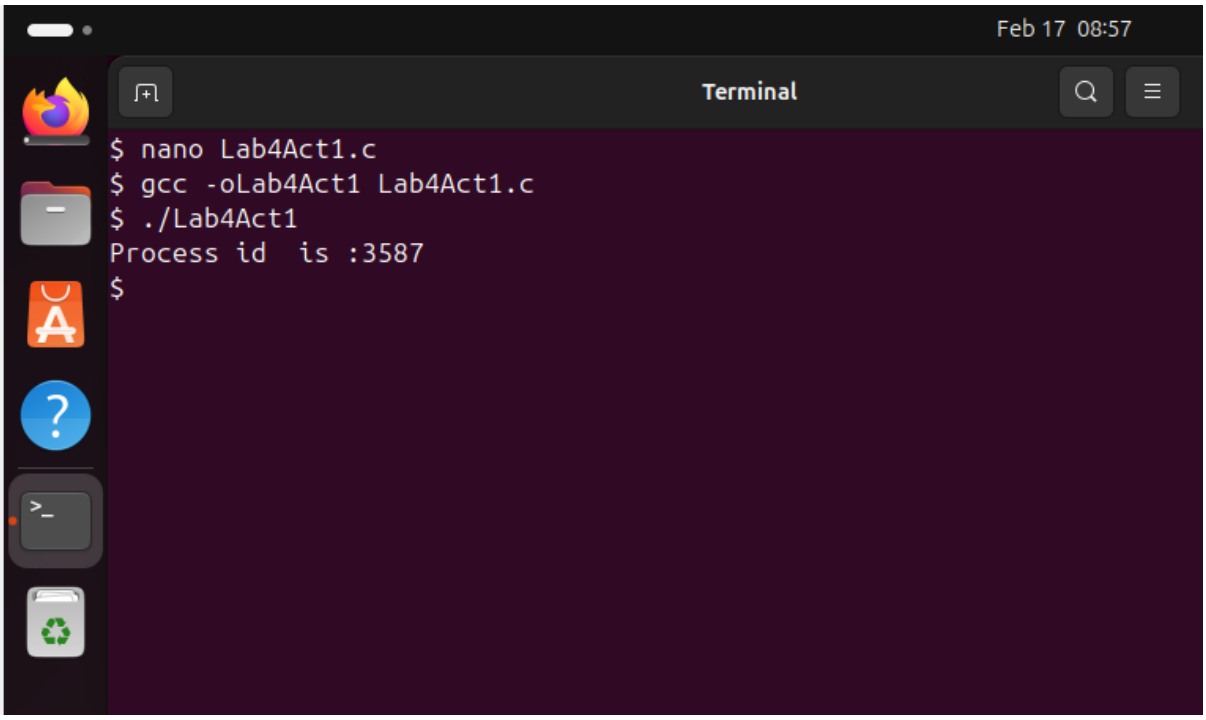
IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 04

Student ID	IT24102555
Student Name	Weerathunga B.A
Campus/ Center name	SLIIT-Malabe
Specialization	Data Science
Batch No	Y2.S2-WD-DS-01.02

Activity 1



a) Process id:3587

b) getpid()

c)It returns the process ID(PID) of the currently running process

d) Compiler warning/error like;

```
$ nano Lab4Act1.c
$ gcc -oLab4Act1 Lab4Act1.c
$ ./Lab4Act1
Process id  is :3587
$ nano Lab4Act1.c
$ gcc -oLab4Act1 Lab4Act1.c
Lab4Act1.c: In function ‘main’:
Lab4Act1.c:4:35: warning: implicit declaration of function ‘getpid’
 [-Wimplicit-function-declaration]
    4 |     printf("Process id  is :%d\n",getpid());
      |                                   ^~~~~~
$
```

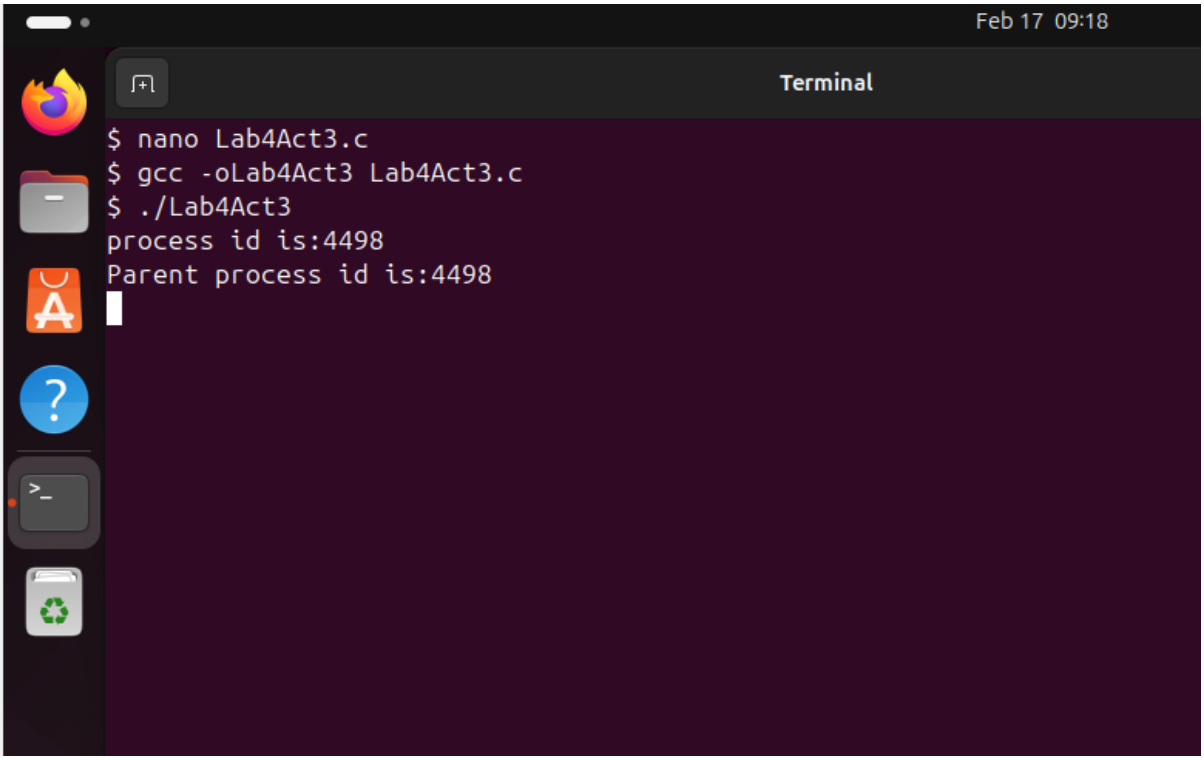
Because getpid() is declare in <unistd.h>

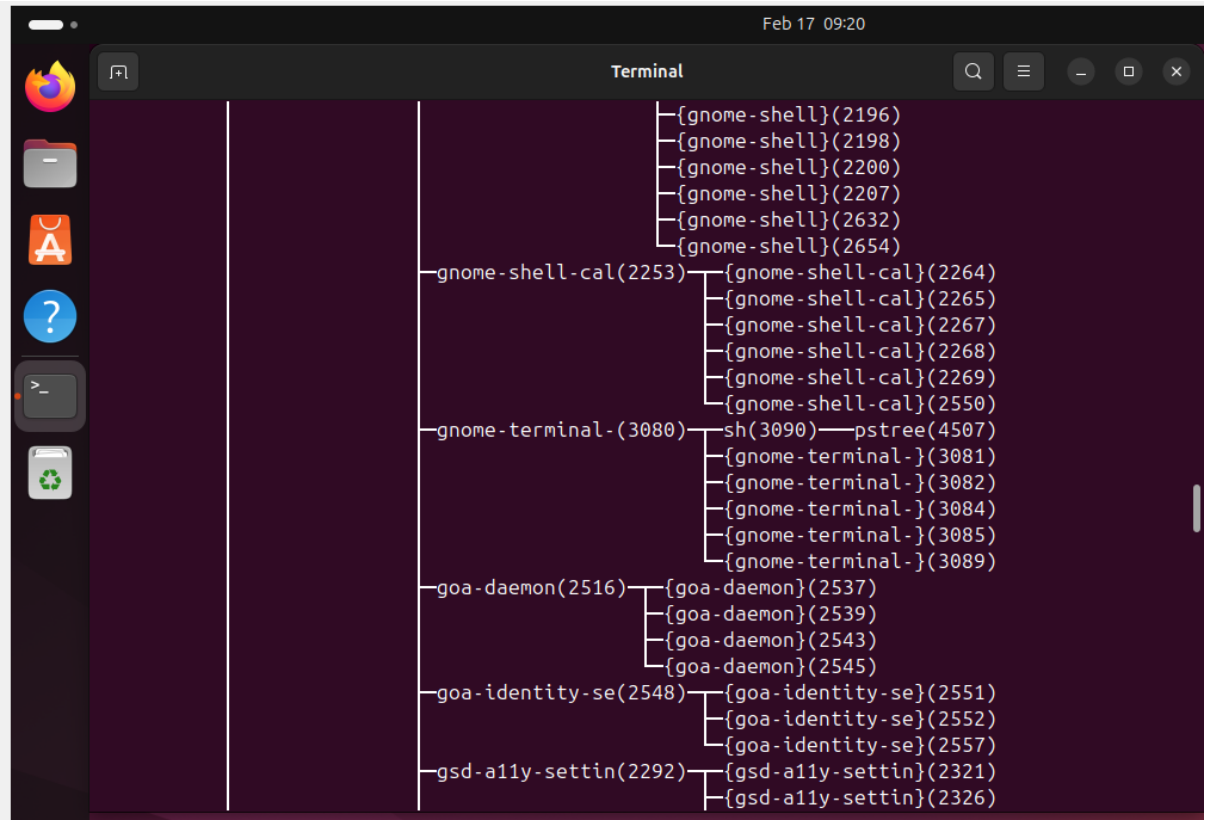
Without it,the complier does not know the function prototype

Activity 2

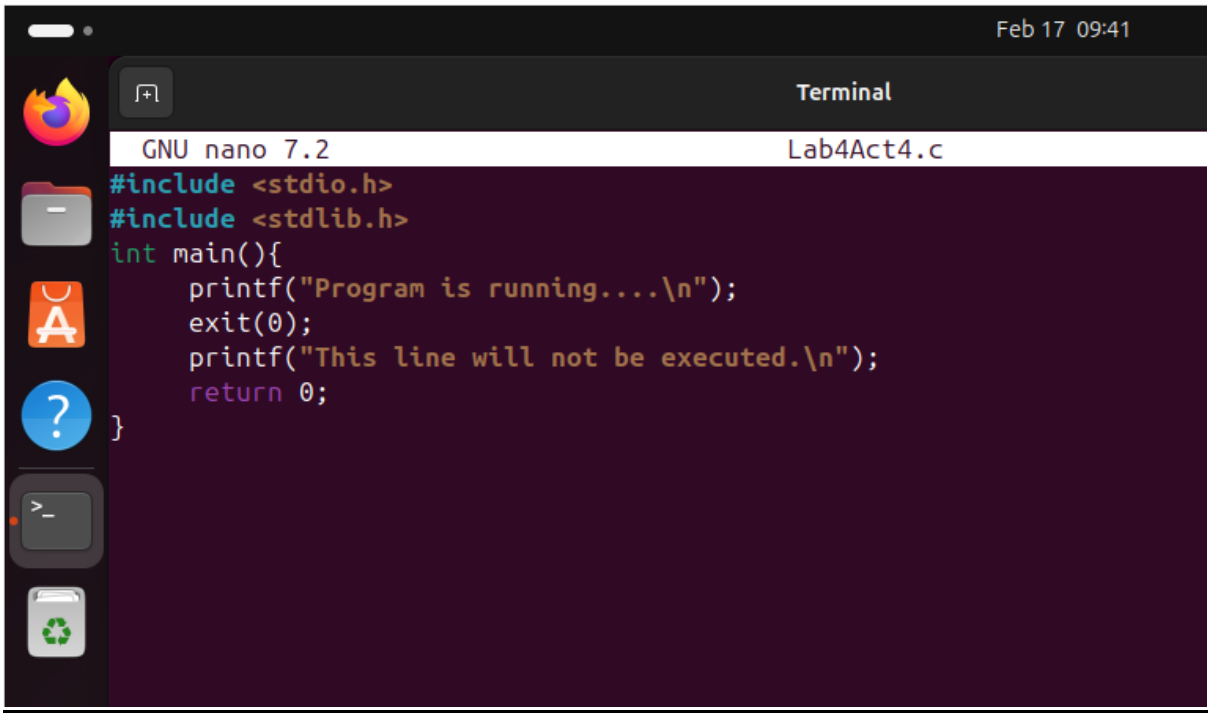
```
$ nano Lab4Act2.c
$ gcc -oLab4Act2 Lab4Act2.c
$ ./Lab4Act2
process id is:4402
Parent process id is:4402
$
```

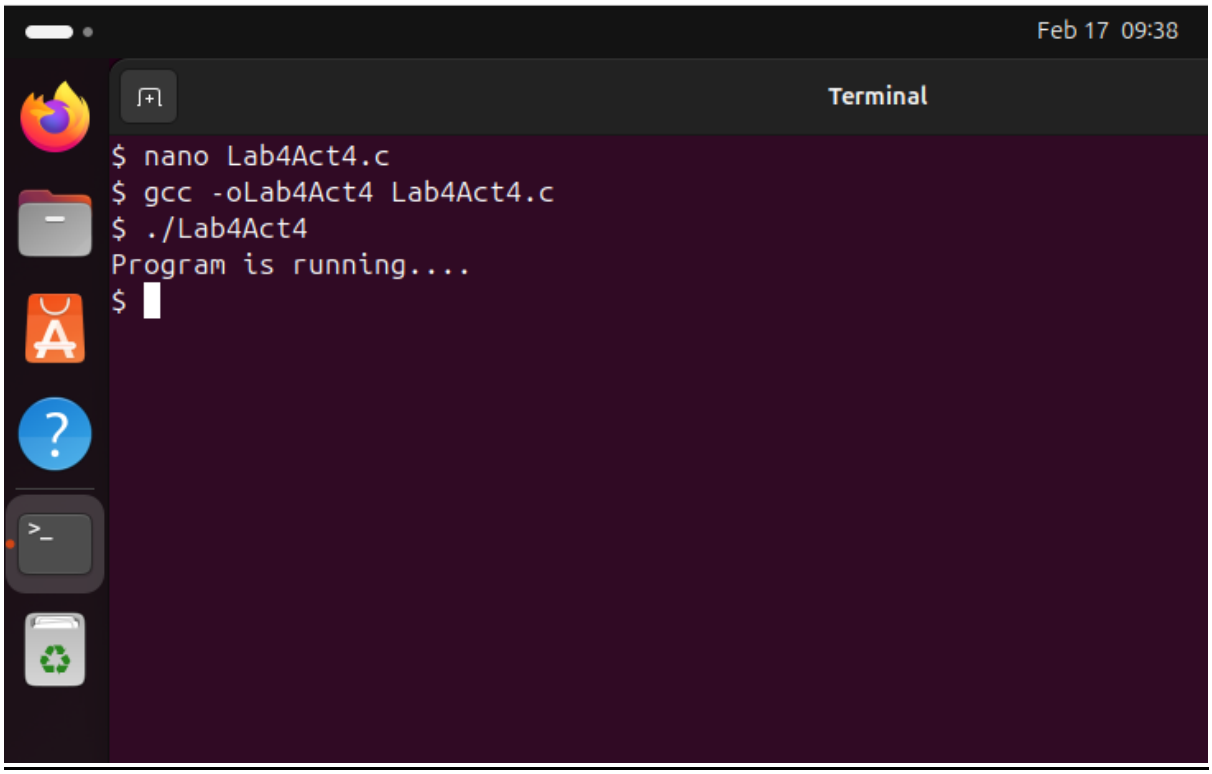
Activity 3


Feb 17 09:18
Terminal
$ nano Lab4Act3.c
$ gcc -oLab4Act3 Lab4Act3.c
$ ./Lab4Act3
process id is:4498
Parent process id is:4498

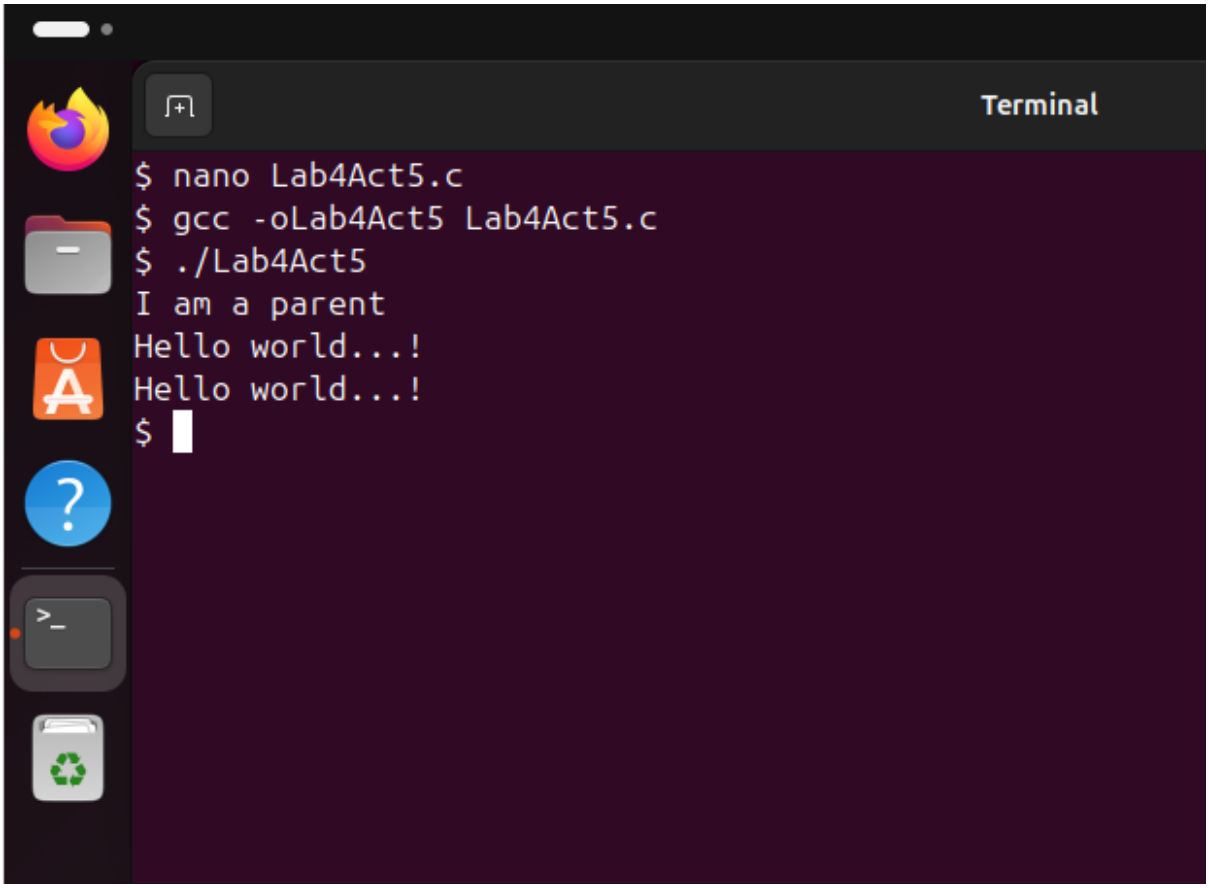

Feb 17 09:20
Terminal
├─{gnome-shell}(2196)
├─{gnome-shell}(2198)
├─{gnome-shell}(2200)
├─{gnome-shell}(2207)
├─{gnome-shell}(2632)
└─{gnome-shell}(2654)
├─gnome-shell-cal(2253)─┬─{gnome-shell-cal}(2264)
├─{gnome-shell-cal}(2265)
├─{gnome-shell-cal}(2267)
├─{gnome-shell-cal}(2268)
├─{gnome-shell-cal}(2269)
└─{gnome-shell-cal}(2550)
├─gnome-terminal-(3080)─┬─sh(3090)──pstree(4507)
├─{gnome-terminal-}(3081)
├─{gnome-terminal-}(3082)
├─{gnome-terminal-}(3084)
├─{gnome-terminal-}(3085)
└─{gnome-terminal-}(3089)
├─goa-daemon(2516)─┬─{goa-daemon}(2537)
├─{goa-daemon}(2539)
├─{goa-daemon}(2543)
└─{goa-daemon}(2545)
├─goa-identity-se(2548)─┬─{goa-identity-se}(2551)
├─{goa-identity-se}(2552)
└─{goa-identity-se}(2557)
├─gsd-a11y-settin(2292)─┬─{gsd-a11y-settin}(2321)
├─{gsd-a11y-settin}(2326)

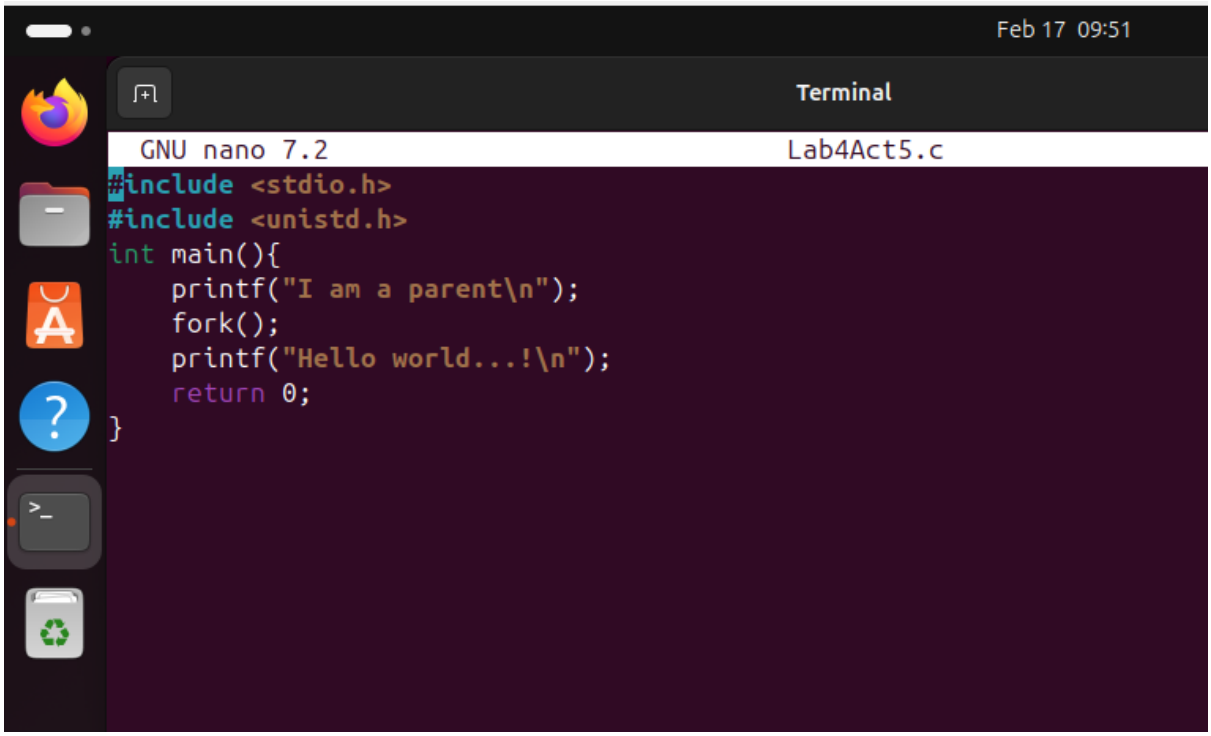
Activity 4


Feb 17 09:41
Terminal
GNU nano 7.2
Lab4Act4.c
#include <stdio.h>
#include <stdlib.h>
int main(){
printf("Program is running....\n");
exit(0);
printf("This line will not be executed.\n");
return 0;
}

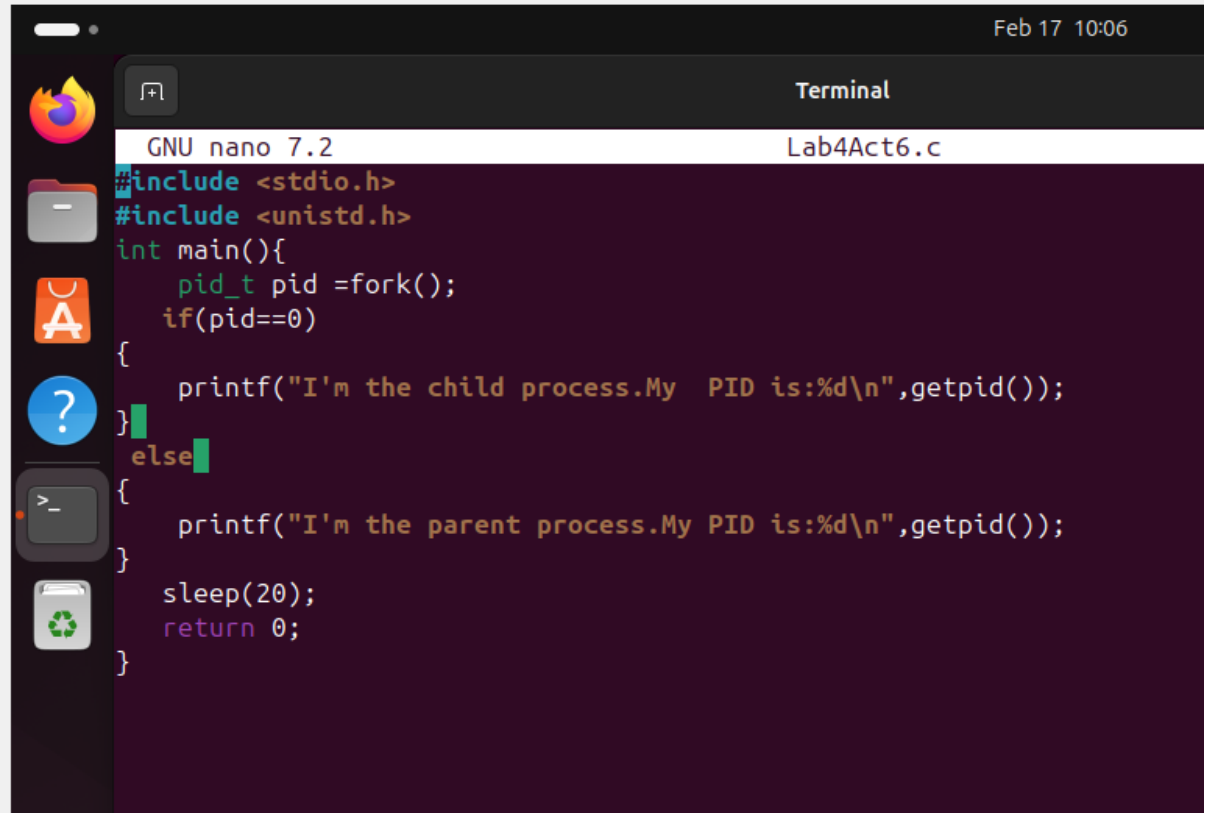

Feb 17 09:38
Terminal
$ nano Lab4Act4.c
$ gcc -oLab4Act4 Lab4Act4.c
$ ./Lab4Act4
Program is running....
$

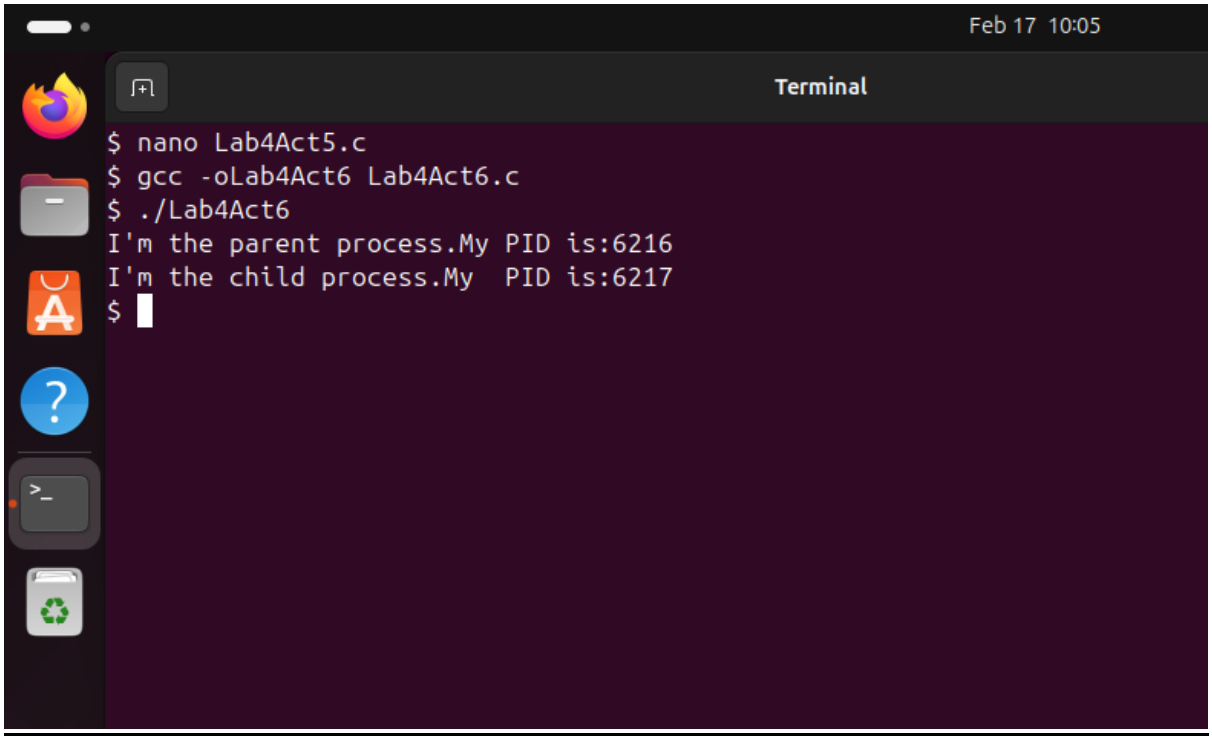
Activity 5


Terminal
$ nano Lab4Act5.c
$ gcc -oLab4Act5 Lab4Act5.c
$ ./Lab4Act5
I am a parent
Hello world...!
Hello world...!
$

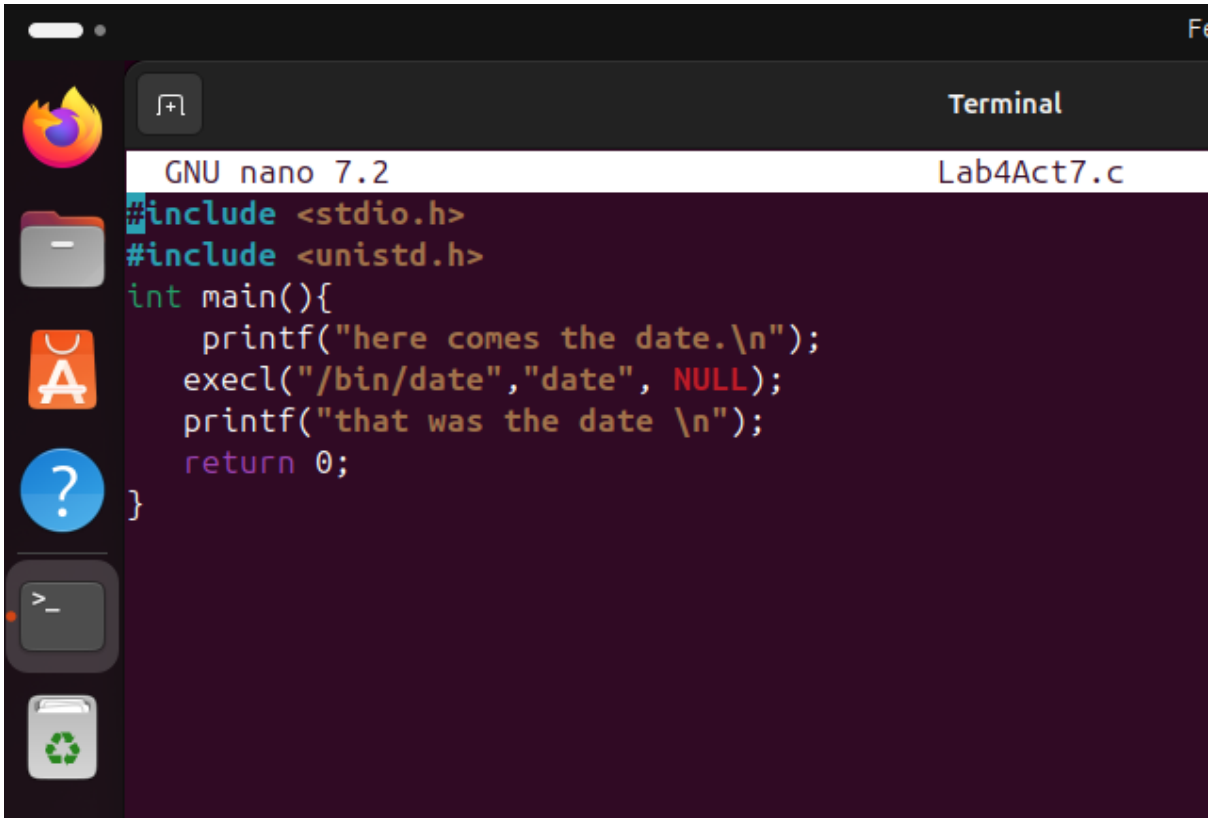

Feb 17 09:51
Terminal
GNU nano 7.2 Lab4Act5.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("I am a parent\n");
fork();
printf("Hello world...!\n");
return 0;
}

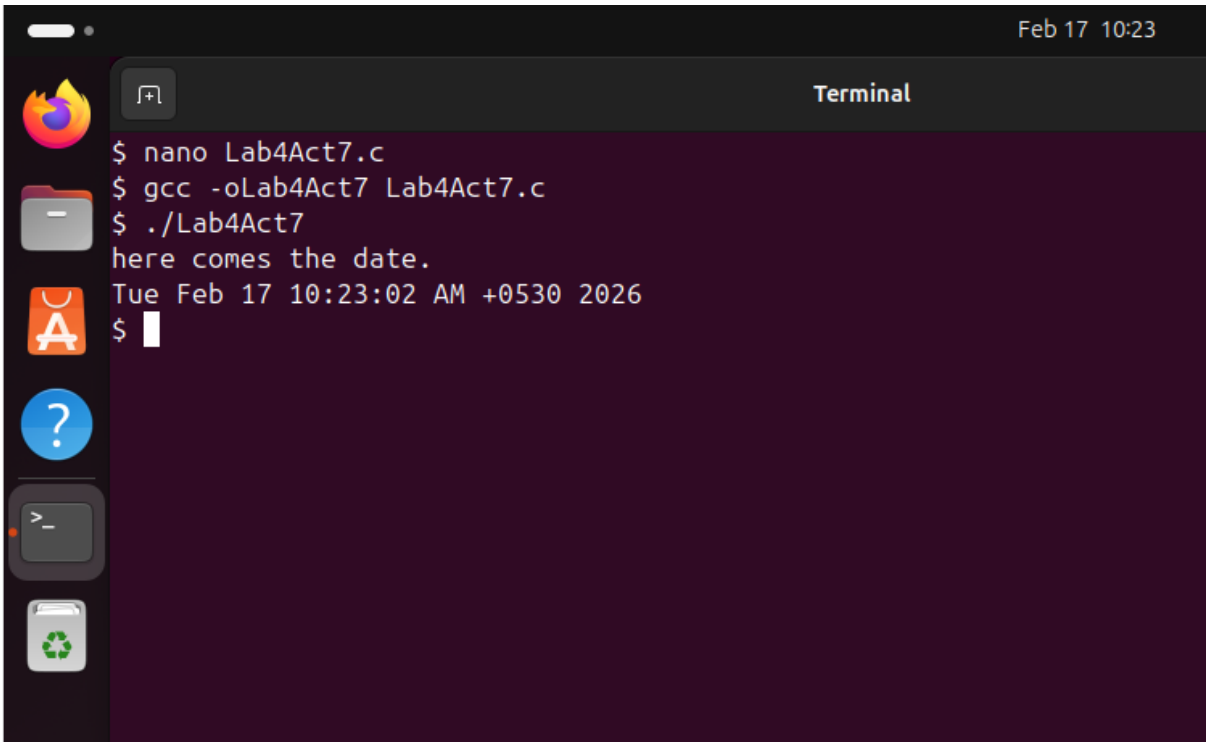
Activity 6


Feb 17 10:06
Terminal
GNU nano 7.2
Lab4Act6.c
#include <stdio.h>
#include <unistd.h>
int main(){
pid_t pid =fork();
if(pid==0)
{
printf("I'm the child process.My PID is:%d\n",getpid());
}
else
{
printf("I'm the parent process.My PID is:%d\n",getpid());
}
sleep(20);
return 0;
}

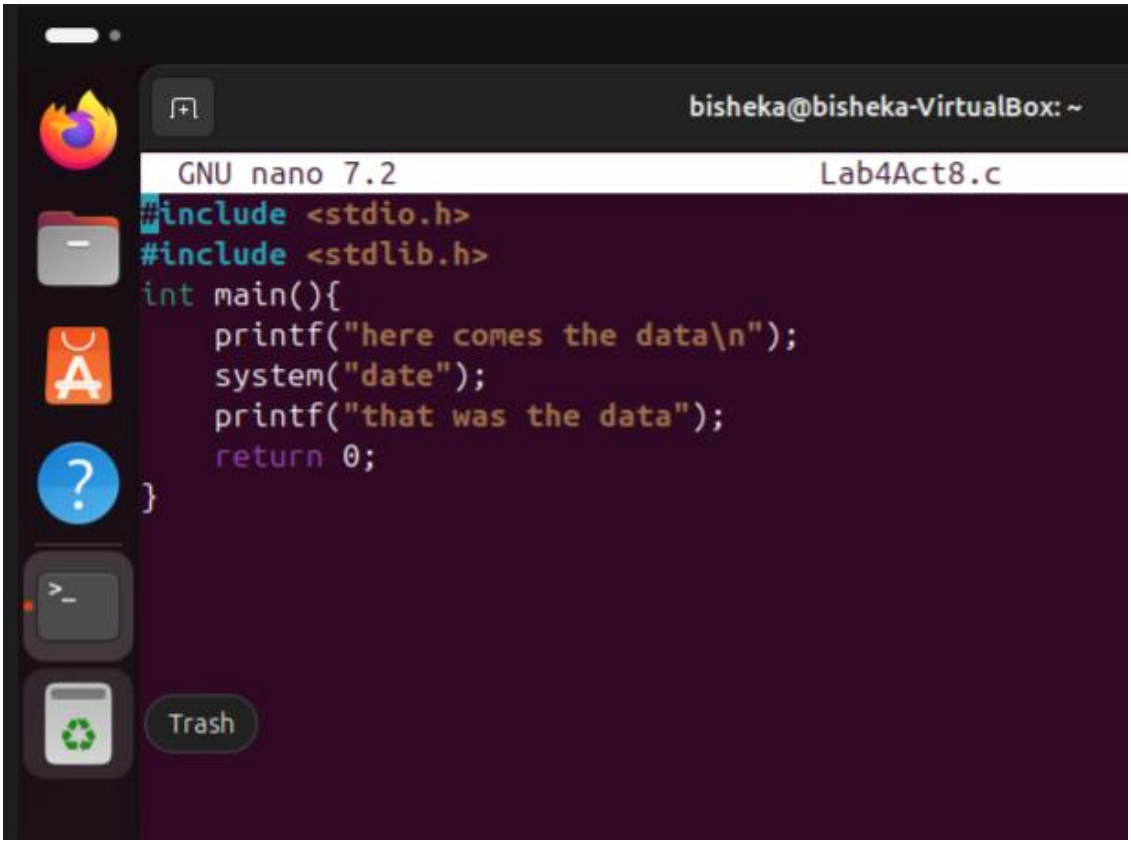

Feb 17 10:05
Terminal
$ nano Lab4Act5.c
$ gcc -oLab4Act6 Lab4Act6.c
$ ./Lab4Act6
I'm the parent process.My PID is:6216
I'm the child process.My PID is:6217
$

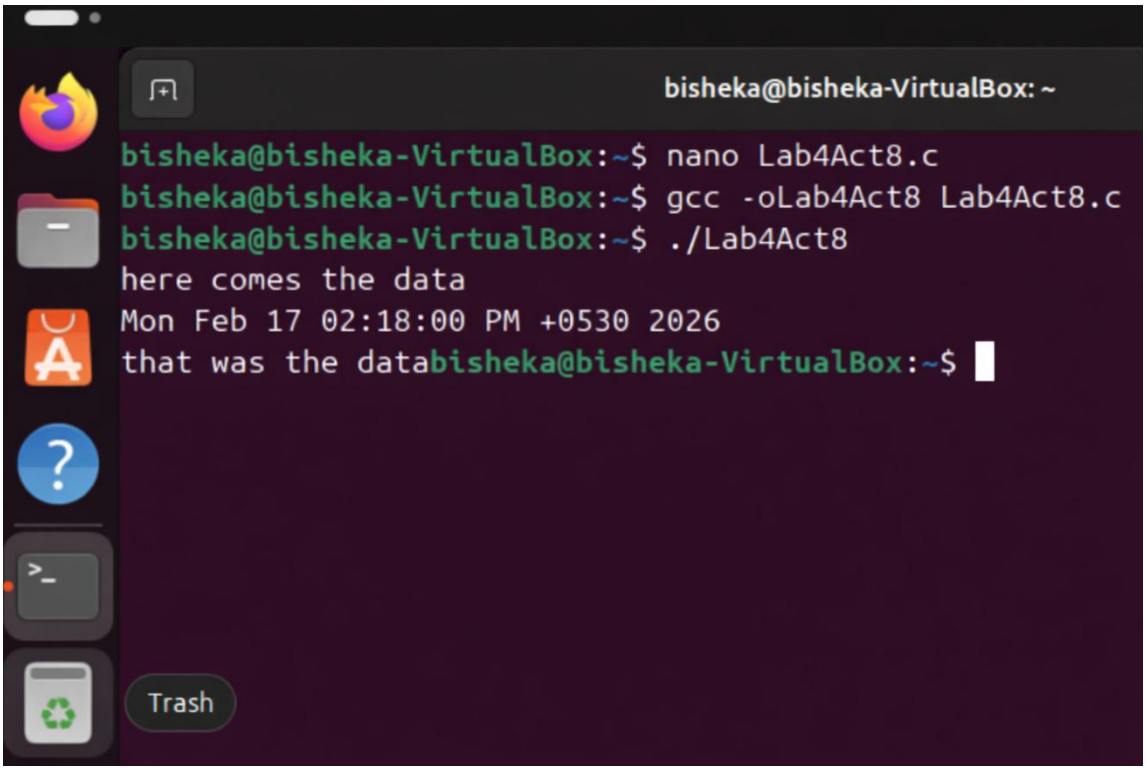
Activity 7


Terminal
GNU nano 7.2
Lab4Act7.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("here comes the date.\n");
execl("/bin/date","date", NULL);
printf("that was the date \n");
return 0;
}


Feb 17 10:23
Terminal
$ nano Lab4Act7.c
$ gcc -oLab4Act7 Lab4Act7.c
$ ./Lab4Act7
here comes the date.
Tue Feb 17 10:23:02 AM +0530 2026
$

Activity 8


bisheka@bisheka-VirtualBox: ~
GNU nano 7.2
Lab4Act8.c
#include <stdio.h>
#include <stdlib.h>
int main(){
printf("here comes the data\n");
system("date");
printf("that was the data");
return 0;
}
Trash


bisheka@bisheka-VirtualBox: ~
bisheka@bisheka-VirtualBox:~$ nano Lab4Act8.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab4Act8 Lab4Act8.c
bisheka@bisheka-VirtualBox:~$ ./Lab4Act8
here comes the data
Mon Feb 17 02:18:00 PM +0530 2026
that was the databisheka@bisheka-VirtualBox:~$
Trash

bisheka@bisheka-VirtualBox: ~

GNU nano 7.2 Lab4Act8.c *

```c
#include <stdio.h>
#include <stdlib.h>
int main(){
    printf("here comes the data\n");
    fork();
    system("date");
    printf("that was the data");
    return 0;
}
```

bisheka@bisheka-VirtualBox: ~

```
bisheka@bisheka-VirtualBox:~$ nano Lab4Act8.c
bisheka@bisheka-VirtualBox:~$ ./Lab4Act8
here comes the data
Mon Feb 17 02:28:58 PM +0530 2026
Mon Feb 17 02:28:58 PM +0530 2026
that was the data that was the databisheka@bisheka-Virtu
bisheka@bisheka-VirtualBox:~$
```

Trash